Aneesh Grover

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Experience

EventPin, Inc. **Sept 2026 – Jan 2027**
Software Engineering Intern *Holland, OH*
- Backend intern building features and improving performance on a live event-discovery platform.

AcademicAlly **July 2024 – August 2025**
Developer – ACM Thapar Society *React, Python, Pandas, BeautifulSoup*
- Built a responsive timetable portal serving **10,000+ students** for instant class schedule lookup across campus.
- Wrote ETL pipelines processing **50,000+ Excel rows** with custom error handling, reducing parsing failures by **95%**.
- Deployed and maintained the portal through its **live run**.

Projects

folio.mcp - Personal MCP Server TypeScript, Cloudflare Workers, Hono, MCP SDK
- Deployed a remote MCP server on Cloudflare Workers exposing **4 tools** covering GitHub activity, portfolio data, and an LLM-invocable contact tool; **live in production**.
- Configured a KV layer for GitHub response caching with TTLs and IP-based rate limiting on the contact tool to prevent abuse.
- Shipped a companion frontend at **mcp.aneeshgrover.me** with dynamic tool discovery; automated deploys via GitHub Actions.

FocusBoard - Whiteboard Task Capture System Kotlin, FastAPI, CRAFT, TrOCR, Phi-3 Mini, TFLite
- Developed the Android app in Kotlin with ML Kit document scanning and an experimental on-device **YOLO/TFLite** whiteboard detector chosen for mobile size constraints.
- Engineered a **3-stage OCR pipeline** using CRAFT for line segmentation, TrOCR for transcription, and Phi-3 Mini for structured calendar JSON generation from handwritten input.
- Introduced per-device request queues to prevent race conditions under concurrent load.

Campus-Compute - Distributed Compute Framework Python, FastAPI, WebSockets, Docker
- Created a LAN-distributed compute system with a centralized coordinator and worker nodes, with automatic discovery and reconnect.
- Designed a capability-aware scheduler using CPU, RAM, and GPU metadata to place tasks across nodes.
- Delivered a real-time WebSocket dashboard for cluster telemetry and job orchestration; **functional**.

Orbital - Cloud IDE Control Plane Go, Docker, JWT, REST APIs
- Architected a control plane in Go that provisions isolated VS Code workspaces on demand via a Dockerized provisioner.
- Implemented a reconciler loop to recover from provisioning failures without manual intervention.
- Secured the API with JWT authentication and a layered package structure separating auth, storage, and orchestration. **In active development**.

Skills

Languages: Python, Go, Java, C++, JavaScript, TypeScript, SQL
Interfaces for Agents: Model Context Protocol (MCP), LLM tool integration, Cloudflare Workers, Hono, REST APIs
Infrastructure for Developers: distributed systems, WebSockets, JWT authentication, system design, Docker, CI/CD, Git
Systems that Learn: PyTorch, TensorFlow, transformers, RAG, vector databases, OpenCV, CUDA
Data & Platforms: Supabase, PostgreSQL, Pandas

Education

University of Toledo **Toledo, OH**
Bachelor of Science in Computer Science and Engineering — GPA: 3.4/4.0 *Expected May 2028*

Leadership & Involvement

University of Toledo Learning Enhancement Center **Sept 2026 – Present**
Computer Science Tutor *Toledo, OH*
- Course tutor for Non-Linear Data Structures, Software Engineering, and DBMS.

Association for Computing Machinery (ACM), Thapar Chapter **Nov 2023 – May 2025**
Executive Board Member & Machine Learning Bootcamp Lead
- Organized Hacklipse 2024, ACM Thapar's hackathon, for **300+ participants**.
- Led ACM's Machine Learning Bootcamp curriculum, mentoring **100+ students** weekly.

Achievements

- Secured **runner-up** in **Open Innovation Category** at **NVIDIA's SatHack**, competing against **1500+ participants**.
- Awarded Best Use Case for AcademicAlly at ACM Projectathon 2024.